MIL-T-43702A

21 August 1975
SUPERSEDING
MIL-T-43702
7 May 1970

MILITARY SPECIFICATION

TIES, RIBBON, TWIST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wire reinforced, paper or thermo-plastic covered twist ribbon ties for closing bags of food products.

1.2 Classification.

1.2.1 Color. Ribbon ties covered by this specification shall be of the following colors, as specified (see 6.2):

White
Yellow
Green
Red
Blue
Black
Orange

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

FSC 8135

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STANDARDS

FEDERAL

FED-STD-595 - Colors
FED-STD-101 - Preservation, Packaging, and Packing Materials: Test Procedures

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

* 3.1 Standard product. The twist ribbon ties delivered under this specification shall be the supplier's commercial product except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including all parts and components, shall be new and interchangeable.

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3.2 Materials. Materials not definitely specified herein shall be of the quality normally used by the supplier of the twist ribbon ties, provided that the complete item complies with all provisions of this specification.

3.3 Design and construction. The twist ribbon ties shall consist of either a paper or thermo-plastic cover, with single-wire reinforcement. Thermo-plastic covers, at the option of the manufacturer, shall be laminated either by the application of an adhesive, or heat sealed, or extruded with a single strand of wire. The wire in the ribbon ties shall be a soft, carbon steel wire, 26 gauge (0.018 inch diameter), so that the tie, after bending, shall have no spring-back in excess of 1/8 inch when tested as specified in 4.3.1. The twist ribbon tie cover shall not separate to expose the ends of the wire and the wire shall not break when the tie is twisted and tested as specified in 4.3.2. The ties shall not delaminate or crack when tested as specified in 4.3.3 and 4.3.4, respectively.

3.3.1 Thickness. The thickness of twist ribbon tie covers, exclusive of the reinforcing wire, shall be 0.007 inch, with a tolerance of plus or minus 0.002 inch.

3.3.2 Width. The width of the twist ribbon ties shall be not less than 9/64 inch nor more than 13/64 inch.

3.3.3 Color. The colors of twist ribbon ties, as specified in 1.2.1, shall conform to any shade of the basic color of FED-STD-595.

3.4 Put-up. The twist ribbon ties shall be oscillate-wound on a plastic spool, with no kinks or twists in the ribbon. The spool shall be wound with a twist ribbon tie of one color, in a continuous length of not less than 2500 feet. The spool shall have a center mounting hole, 1-1/16 inch diameter, plus or minus 1/32 inch. The spool shall have provisions for securing the free end of the ribbon tie to prevent unravelling.

3.5 Identification. The spool shall have an identification label including the following information:

(a) Name of supplier
(b) Nomenclature and color
(c) Number of feet of twist ribbon tie on spool (not less than 2500 feet)

The lettering shall be legible. The label and letter size and color of lettering shall be at the supplier's option.

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3.6 Workmanship. The twist ribbon ties shall be smooth, free of blisters, cuts or tears and the cover shall not be delaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

* 4.2.1 Component and material inspection. In accordance with 4.1, above, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2.2 Examination of the end item. The lot shall consist of all spools of twist ribbon ties of one color offered for inspection at one time. The sample unit shall be one spool of twist ribbon tie.

4.2.2.1 Visual examination. Examination of the twist ribbon ties shall be made in accordance with classification of defects in table I, as applicable. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units. Examination of the twist tie for defects listed in table I shall be made on eight consecutive yards of the spool, selected at random.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Design and construction	Not as specified	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Spool	Wound ribbon tie not free of kinks or twists in ribbon	X	
	Ribbon color not as specified		X
	Provision to prevent unravelling of spool missing	X	
	Center hole for mounting missing	X	
	Spool not oscillate-wound		X
Label	Not affixed to spool		X
	Does not contain required information		X
Workmanship	Ribbon tie not smooth, free of cuts, blisters or tears	X	
	Tie cover delaminated	X	

4.2.2.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional and continuity requirements. The complete spool shall be unrolled and the thickness shall be measured at the beginning and the end and at the approximate midpoint of the continuous length. Any nonconforming dimension or discontinuity shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. Tests shall be performed on ties as specified in 4.3.1, 4.3.2, 4.3.3 and 4.3.4. The inspection level shall be S-1 with an AQL of 2.5, expressed in terms of defects per hundred units.

* 4.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine whether the packing and marking complies with the section 5 requirements. Defects shall be as listed in table II. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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TABLE II. Examination of preparation for delivery requirements

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as bulging or distortion of containers
Quantity	Not as specified

* 4.2.3.1 Examination of shipping container. When shipping containers are required to comply with PPP-B-636, examination for defects in the closure, waterproofing and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.3 Tests.

4.3.1 Bending test. The flat side of the twist ribbon tie from a sample spool shall be held firmly by the thumb as close as possible to the edge of a nominal 2 X 4 inch by no more than 4 inches long block of wood so that the free end of the tie shall extend 1 inch, plus or minus 1/32 inch, beyond an adjacent right angle surface of the block. The free end of the tie shall be bent flat on the adjacent surface, freed of any restraint and then checked to determine compliance with the bending requirements of 3.3. Any springback of the free end of the tie in excess of 1/8 inch shall constitute failure of this test.

4.3.2 Twist test. A test specimen of twist ribbon tie 5 inches long shall be cut from a sample spool. The specimen shall be wrapped around a wooden dowel, approximately 3/4 inch diameter by 5 inches long, with the free ends of the twist ribbon tie at equal lengths. The tie shall be twisted in four 180 degree turns to form a tight band around the dowel. The twisted ends shall then be checked to determine compliance with the twisting requirements of 3.3. Any nonconformance shall constitute failure of this test.

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4.3.3 Water resistance test. A test specimen of twist ribbon tie approximately 5 inches long shall be cut from a sample spool. The specimen shall be tested for 24 hours in accordance with procedure A of Test Method 3028 of FED-STD-101. Failure to meet the requirements of 3.3 shall constitute failure of the test.

4.3.4 Cold test. A test specimen of twist ribbon tie approximately 5 inches long shall be cut from a sample spool. The specimen shall be subjected to a temperature of minus 20 ± 5 F for 24 hours and while at this temperature shall be tested as specified in 4.3.1. Failure to meet the requirements of 3.3 shall constitute failure of the test.

5. PREPARATION FOR DELIVERY

* 5.1 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.1.1 Level A. Six spools of ties of one color only shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC grade V2s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

* 5.1.2 Level B. Six spools of ties of one color only shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

* 5.1.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

* 5.1.3 Level C (commercial packing). Spools of ties shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification as applicable.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The twist ribbon ties covered herein are for use with automatic and semiautomatic machinery for closing bags filled with bakery, produce, or fruit products weighing not more than 10 pounds.

6.1.1 The various colors of the twist ribbon ties are intended for use to indicate the different days of the week on which the contents were bagged.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color required (see 1.2).
(c) Selection of applicable levels of packing (see 5.1).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.1.1).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

Custodians:

Army - GL
Navy - SA
Air Force - 69

Review activities:

Army - MD, EA
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Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER AND TITLE

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NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

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